# Jiahua Zhang

Incoming MEng in Electrical & Computer Engineering, National University of Singapore · Jan 2027 – 2028 (expected)

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## EDUCATION

**National University of Singapore** Jan 2027 – 2028 (expected)

MEng, Department of Electrical and Computer Engineering · Supervised by Prof. Changsheng Wu

**Nanyang Technological University, Singapore** Aug 2022 – Jun 2026

B.Comp. (Honours) in Computer Science, College of Computing and Data Science

- Specialisation in Data Science.
- Five papers published as an undergraduate, including a first-authored CCF-A paper; research carried out at Guangzhou National Laboratory and several universities.

## PUBLICATIONS

1. **Jiahua Zhang**, Y. Tian, J. Liang, D. Lin, Y. Cai, Y. Liu, Z. Huang, J. Ni, J. He. Process Reward Model as Cross-Task Learner: An Extension of Vision-Language Model to Zero-Shot Medical Reasoning. *Proceedings of the 34th ACM International Conference on Multimedia (ACM MM)*, 2026.
2. **Jiahua Zhang**, J. Liang, Y. Cai, J. Ni, J. He. Can LLM Understand Medical Imaging with Long Context? *IEEE International Conference on Multimedia and Expo (ICME)*, 2026. ORAL
3. **Jiahua Zhang**, Y. Tian. DiffiT-HSFDA: Diffusion Based Source-Free Domain Adaptation for Histopathology. *Advanced Intelligent Computing Technology and Applications (ICIC)*, Springer LNCS 15867, pp. 301–311, 2025. ORAL
4. C. Zhang, X. Wang, H. Zhang, **Jiahua Zhang**, H. Zhang, C. Liu, P. Han. LayerLog: Log Sequence Anomaly Detection Based on Hierarchical Semantics. *Applied Soft Computing*, 132:109860, 2023. (IF 7.8, JCR Q1)
5. C. Zhang, X. Wang, **Jiahua Zhang**, S. Li, H. Zhang, C. Liu, P. Han. VESC: A New Variational Autoencoder Based Model for Anomaly Detection. *International Journal of Machine Learning and Cybernetics*, 14(3):683–696, 2022. (IF 2.9)

## RESEARCH EXPERIENCE

**Guangzhou National Laboratory** Feb 2024 – Oct 2026

Research Engineer (May – Oct 2026) · Research Collaborator (Feb 2024 – Apr 2026)

Prof. Jianxing He's group, a joint team of Guangzhou National Laboratory and the State Key Laboratory of Respiratory Disease, Guangzhou Medical University.

- **AI sandbox hospital:** core developer of a simulated hospital in which LLM agents take on clinical roles; under review at ACL ARR.
- **Medical reasoning chains:** wrote the group's specification for medical chain-of-thought — how EHR reasoning is decomposed into steps, how clinical visual evidence is cited, and the principles for RLHF annotation — used as the common reference for multimodal data construction and model evaluation.
- **Reasoning evaluation:** designed the reasoning-chain evaluation framework — dimensions, scoring rules and spot-check protocol — used to compare models and data versions across iterations, and turned it into an annotation protocol for a team of medical students.
- **Process reward modelling (BioMedPRM):** designed a process reward model that scores each step of a biomedical reasoning chain rather than only its final answer and transfers across tasks, with emphasis on robustness under distribution shift and in out-of-distribution clinical settings.
- **Source-free domain adaptation:** found that weakly supervised segmentation of lung-cancer histopathology loses region-of-interest activation when transferred across centres with different staining protocols; designed a diffusion-based source-free adaptation pipeline that restores high-accuracy segmentation without source data or extra annotation.
- **Training-free multimodal LLM:** built a training-free framework for CT and whole-slide images, adapting long-context models to high-dimensional medical imaging without task-specific fine-tuning.
- **Output:** first-authored papers at ACM Multimedia 2026, IEEE ICME 2026 (oral) and ICIC 2025 (oral); one submission under review at ACL ARR.

**Nanyang Technological University — School of Social Sciences (Psychology)** Oct 2024 – May 2025

Research Assistant, supervised by Asst. Prof. Luo Lizhu

- Represented high-dimensional biomedical data — functional MRI, whole-slide images — as graphs so that language models can reason over far more variables than a context window admits.
- Proposed a training-free graph in-context learning method for medical multimodal prediction and benchmarked it against fine-tuned baselines.

**Pohang University of Science and Technology (POSTECH) — DPNM Laboratory** Jun 2024 – Aug 2024

POSTECH Summer Program (PSP), supervised by Prof. James Won-Ki Hong

- Designed a convolutional diffusion-denoising pipeline for MRI and X-ray segmentation and spinal disease classification, improving results on the BUU and lumbar-spine MRI datasets.
- Applied representation learning and NLP to hospital chief-complaint text, evaluating TF-IDF and other vectorisation and clustering methods, and developed lightweight algorithms for edge computing.

**Nanyang Technological University — S-Lab for Advanced Intelligence** Mar 2024 – Jun 2024

Student Research Assistant — LLM-based video generation

- Developed and applied noise-detection algorithms to improve the visual quality of AI-generated video.
- Built the label preprocessing pipeline for a generated-video quality evaluation benchmark.

**Harbin Institute of Technology, Shenzhen** Jan 2022 – Jul 2022

Research Assistant

- Owned data processing for two anomaly-detection studies: cleaning and preprocessing large-scale log and time-series data, and preparing training and evaluation sets.
- Produced the results visualisations for both studies, including the comparative and ablation figures reported in the papers.
- **Output:** two co-authored journal papers.

## OPEN-SOURCE WORK

**Open-Source Speech Synthesis Toolkit** 2022 – 2023

Independent developer

- Trained and released multi-speaker speech synthesis models (VITS / Bert-VITS2) covering a large voice cast.
- Released vits with chatgpt-gpt3, an open-source project chaining VITS synthesis to LLM dialogue so characters can both converse and speak; adopted by content creators for video production and livestreaming (385★ on GitHub).
- Maintained a public demo on Hugging Face Spaces.

## SKILLS

**LLMs & reward modelling:** process reward models (step-level scoring), vision-language models, long-context multimodal reasoning, test-time computation, training-free in-context learning; medical chain-of-thought specification, reasoning-chain evaluation, benchmark and label construction.

**LLM agents:** multi-agent clinical simulation, role-conditioned agents, multi-agent workflow orchestration.

**Generative models:** diffusion models (FLUX / SDXL), LoRA composition and parameter-space analysis (SVD, polar decomposition), domain adaptation and transfer learning, VITS / Bert-VITS2 speech synthesis.

**ML frameworks:** PyTorch, Hugging Face Transformers, PyTorch Geometric, torchvision, torchaudio; large-model training and inference on multi-GPU clusters.

**Programming:** Python, C# / .NET, SQL, Flask, React, Unity.

**Data:** curation and preprocessing pipelines for large-scale clinical, imaging, log and time-series data; clinical annotation protocols.

**Medical & multimodal data:** CT, MRI, X-ray, whole-slide pathology images, functional MRI.

**Languages:** Mandarin Chinese (native), English (IELTS 7.0).

## SERVICE & CAMPUS ACTIVITIES

**NTU Chinese Society — Spring Festival Gala, Committee Member** 2025

Helped organise the stage production for the NTU 2025 Spring Festival Gala, covering logistics and on-stage support.

**NTU Chinese Society — Publicity Team, Committee Member** 2023 – 2024

Event manager in the publicity team; wrote and typeset articles for the society's WeChat account across the academic year.

**NTU IET Club — Event Manager** Aug 2022 – May 2023

Organised IT-themed events and technical competitions for club members and the wider student body; led a committee of five across three events, owning planning, publicity and delivery.

**Shenzhen International High School Students Forum — Student Volunteer** 2019

Supported delegates from different school systems in preparing and presenting their work, and assisted with on-site coordination.